SUPPLIER SPECIFICATION MANUAL

Component: Battery Management System (BMS) Controller
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Classification: Automotive Grade / ASIL-D
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1. Scope

This document defines the technical specifications, interface requirements, and environmental operating limits for the Bosch BMS Controller, hereinafter referred to as "the unit." The unit is designed for integration into high-voltage electric vehicle powertrain architectures.

2. Functional Overview

- Monitors and manages Li-ion battery pack state-of-charge (SoC) and state-of-health (SoH)
- Provides cell-level voltage and temperature monitoring across up to 192 cells in series
- Executes active and passive cell balancing algorithms
- Interfaces with vehicle powertrain control unit (PCU) for high-voltage contactor management
- Compliant with ISO 26262 (ASIL-D) functional safety requirements

3. High-Voltage System Compatibility

- Rated for integration with **400V nominal** high-voltage battery architectures
- Supports pack voltage range of 250V–450V DC
- Isolation resistance monitoring per ISO 6469-3
- Integrated pre-charge circuit control for main contactor engagement

4. Communication Interfaces

- Primary vehicle network interface: **standard CAN (ISO 11898-2)**, 500 kbit/s
- Diagnostic interface: UDS over CAN (ISO 14229)
- Wake-up via CAN network activation line

5. Environmental Limits

The following limits define the certified operating envelope for the unit. Operation outside these limits voids the manufacturer's functional safety certification and warranty coverage.

- Storage temperature range: -40°C to +85°C
- Operating humidity: 5%–95% RH, non-condensing
- IP rating: IP67 (housing, connector-mated)

The absolute maximum operating temperature for this BMS unit is 75°C.

- Thermal derating of balancing current begins at 60°C ambient
- Unit shall not be operated in continuous ambient conditions exceeding the stated maximum without active liquid cooling supplementation

6. Technical Specifications Table

Parameter	Value	Unit	Notes
Nominal Bus Voltage	400	V DC	±50V tolerance
Max Cells Monitored	192	cells	Series configuration
Cell Voltage Accuracy	±2	mV	Over full temp range
Communication Protocol	CAN 2.0B	—	ISO 11898-2
CAN Bus Speed	500	kbit/s	Standard CAN
Max Operating Temperature	75	°C	Absolute maximum
Min Operating Temperature	-40	°C	Cold-crank capable
Ingress Protection	IP67	—	Mated connectors
Functional Safety Rating	ASIL-D	—	ISO 26262
Quiescent Current	<2	mA	Sleep mode

7. Compliance & Certification

• ISO 26262:2018 (ASIL-D)

• ISO 6469-3 (Electrical safety)

• IATF 16949 manufacturing certification